Cost-evolution programme for formulation and production

Starting problem

In the industrial environment there is a clear need for a centralised, traceable and operationally robust tool to continuously analyse formulation and production cost evolution. When this information is fragmented across multiple files, without structured version control or agile time-based reading, technical and economic decision-making becomes harder.

Proposed solution

The proposal is a Streamlit-based web application that integrates historical cost data into a single environment and allows it to be treated, visualised and analysed dynamically.

The tool makes it possible to monitor cost evolution over time, identify relevant deviations and analyse the impact of modifications in formulation or production conditions.

Improvement goals

- Reduce cost variability through structured historical analysis.
- Improve formulation decisions by relying on real data and trends.
- Minimise unnecessary adjustments through better understanding of system behaviour.
- Increase operational robustness and traceability, aligning with demanding industrial and quality-audit environments.

Underlying reading

The value of the tool is not only in data visualisation. It lies in creating a technical base that allows the organisation to move from scattered monitoring to a continuous and useful reading of cost evolution, deviations and their relationship with formula and process changes.

System screenshots

Evolucion de Costes

Carga automatica de Excel/CSV desde una carpeta configurable.

Carpeta de entrada

./data

Filtros iniciales

Pintura (descrip_pt)

B-200 MATE SUP… ×

Color (

BLA

Fecha inicio

2019/01/18

Fecha fin

2026/03/17

Ficheros detectados: 2 | Registros cargados: 29415 | Registros tras filtros: 56

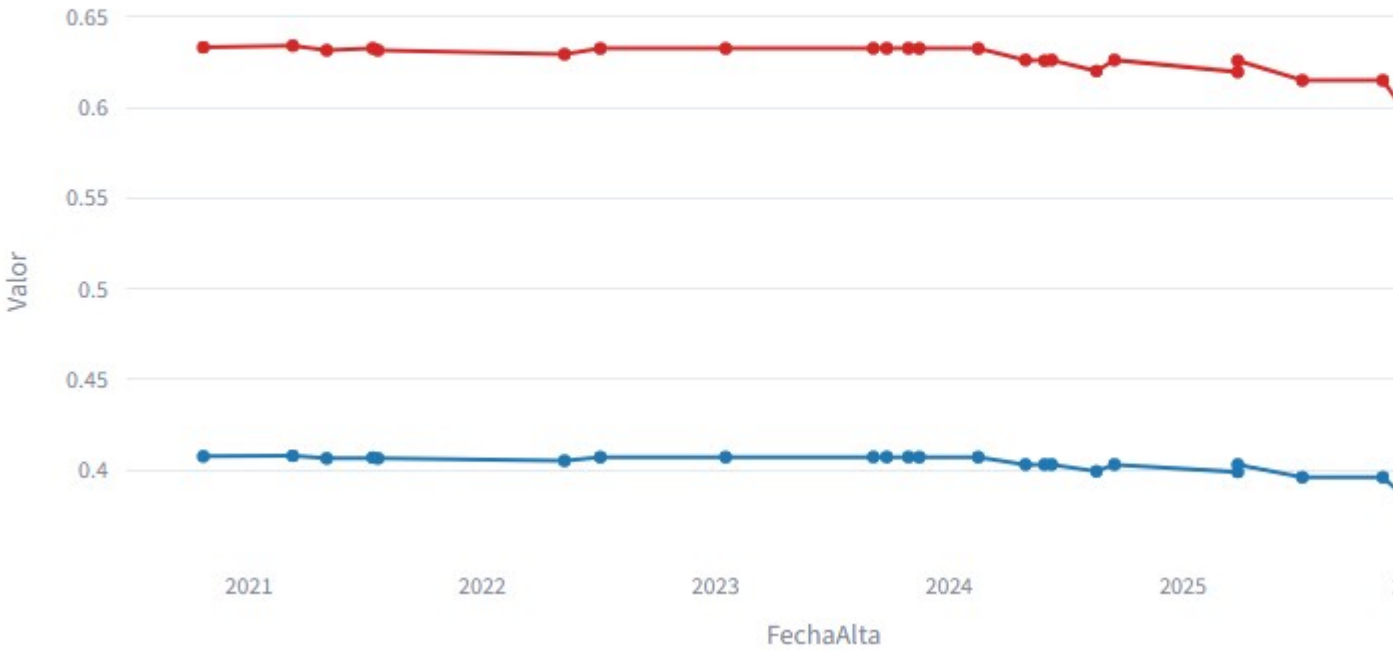
Visualizacion operativa

Vista operativa de costes (inspirada en Coste Productos)

1 registros en esta consulta

IdPT	Articulo	Color	Cos
PT1692	B-200 MATE SUPER +	BLANCO	

Evolucion temporal: Euros/Kg y Euros/Lt



Variacion porcentual respecto al primer registro

Alertas de densidad (>10% de descenso)

No se detectaron descensos bruscos de densidad superiores al 10%.

Dataset filtrado

	id_fase	fecha_alta	id_pt	descrip_pt	p1
0	25136	2020-10-22	PT1692	B-200 MATE SUPER +	361
1	25136	2020-10-22	PT1692	B-200 MATE SUPER +	361
2	25623	2021-03-11	PT1692	B-200 MATE SUPER +	361
3	25623	2021-03-11	PT1692	B-200 MATE SUPER +	361
4	25814	2021-05-03	PT1692	B-200 MATE SUPER +	361
5	25814	2021-05-03	PT1692	B-200 MATE SUPER +	361
6	26740	2021-07-14	PT1692	B-200 MATE SUPER +	361
7	26740	2021-07-14	PT1692	B-200 MATE SUPER +	361
8	26820	2021-07-22	PT1692	B-200 MATE SUPER +	361

Descargar CSV del dataset filtrado